

Chapter 7: Terms of Trade

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Outline

- 1 Introduction
- 2 Terms of Trade and Trade Balance
- 3 Terms of Trade and Business Cycle
- 4 The MX Model
- 5 Extension: Alternative Detrending Methods [New]
- 6 References

Chapter 7 – Replication and Extension

- We **replicate the empirical analysis** in Chapter 7.
- The replication is conducted in Python and reproduces the results reported in the book.
- We then extend the analysis to assess the robustness of the results to alternative detrending methods.

Extension: Does the detrending method matter?

We compare the authors' **quadratic detrending** with:

- Hodrick–Prescott (HP) filter;
 - Hamilton filter.
-
- This extension is motivated by Carlos Eduardo's comment that the choice of detrending method may affect the results.

Motivation: Why Terms of Trade Matter?

- The **Terms of Trade (ToT)** capture the relative price of a country's exports and imports:

$$ToT_t = \frac{P_t^X}{P_t^M}.$$

- For emerging economies, ToT fluctuations can generate large changes in **national income and purchasing power**.
- A deterioration in the ToT reduces the amount of imports that can be purchased with a given volume of exports.
- As a result, ToT shocks can propagate to key macroeconomic variables such as **output, consumption, investment, and the trade balance**.

Key Motivation

How important are Terms-of-Trade shocks for macroeconomic fluctuations?

From Terms of Trade to the MX Model

1 Terms of Trade

► Terms of Trade and the Trade Balance

- We study how ToT shocks affect the **trade balance**.
- We estimate the persistence of ToT fluctuations and analyze the dynamic response of the trade balance.
- We also test whether the response depends on the **persistence of ToT shocks**.

► Terms of Trade and Business Cycles

- We extend the analysis to **output, consumption, investment, and the trade balance**.
- We quantify how these variables respond to ToT shocks.
- We estimate the share of macroeconomic fluctuations **explained by Terms-of-Trade shocks**.

2 MX Model

- A **small open-economy model with incomplete markets** used to quantify the macroeconomic effects of ToT shocks.
- We use the estimated ToT process to **parameterize the model**.
- We evaluate how ToT shocks propagate to **output, consumption, investment, and the trade balance**.

3 Robustness to Detrending [Extra Work]

- We test whether the results are sensitive to the **detrending method**, comparing quadratic detrending with the HP and Hamilton filters.

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Terms of Trade and the Trade Balance

Empirical Question

Does an improvement in the Terms of Trade cause an **improvement or deterioration of the trade balance**?

- We estimate a country-specific SVAR for the **Terms of Trade** and the **trade balance**. More on Terms of Trade
- The cyclical components are obtained using **quadratic detrending**.¹

$$\begin{aligned}\widehat{tot}_t &= \rho \widehat{tot}_{t-1} + \pi \varepsilon_t^1, \\ \widehat{tb}_t &= \alpha_0 \widehat{tot}_t + \alpha_1 \widehat{tot}_{t-1} + \rho_2 \widehat{tb}_{t-1} + \sqrt{\sigma_{22}} \varepsilon_t^2.\end{aligned}$$

- We identify ε_t^1 as the **Terms-of-Trade shock**:
 - ▶ ToT is assumed to be contemporaneously unaffected by other shocks;

$$\begin{bmatrix} \widehat{tot}_t \\ \widehat{tb}_t \end{bmatrix} = h_x \begin{bmatrix} \widehat{tot}_{t-1} \\ \widehat{tb}_{t-1} \end{bmatrix} + \Pi \begin{bmatrix} \varepsilon_t^1 \\ \varepsilon_t^2 \end{bmatrix}, \quad h_x \equiv \begin{bmatrix} \rho & 0 \\ \alpha_0 \rho + \alpha_1 & \rho_2 \end{bmatrix}, \quad \Pi \equiv \begin{bmatrix} \pi & 0 \\ \alpha_0 \pi & \sqrt{\sigma_{22}} \end{bmatrix}.$$

¹Later, we assess robustness using alternative detrending methods.

How Should the Trade Balance Respond to ToT Shocks?

- A positive ToT shock may improve the trade balance, but the magnitude of the response depends on the underlying mechanism.

1. Harberger–Laursen–Metzler (HLM)

A ToT improvement raises real income and savings, improving the trade balance.

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \alpha}{1 + \mu - \alpha} q > 0$$

Prediction: a positive ToT shock improves the trade balance.

Derivation

2. Obstfeld–Svensson–Razin (ORS)

The response depends on the **persistence** of the ToT shock:

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \rho}{1 + r - \rho} \quad \rho \uparrow \Rightarrow \Delta tb \downarrow$$

Prediction: more persistent shocks generate a smaller improvement.

Derivation

Testing the ORS Effect

Testable Prediction

The ORS effect predicts that countries with more persistent ToT shocks should exhibit a **smaller trade-balance response**.

From the country-specific SVAR, for each country i we recover:

$$\underbrace{\rho_i}_{\text{Persistence of ToT shocks}} \quad \text{and} \quad \underbrace{\frac{\pi_{21,i}}{\pi_{11,i}} \times 10}_{\text{TB impact response to a 10\% ToT shock}}.$$

The ORS hypothesis therefore implies:

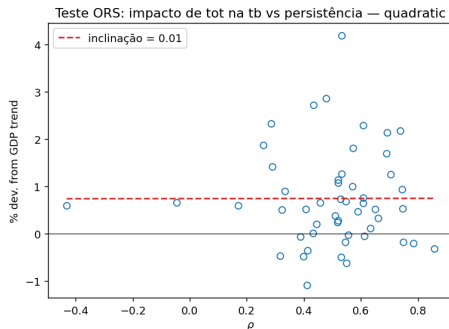
$$\boxed{\rho_i \uparrow \implies \frac{\pi_{21,i}}{\pi_{11,i}} \times 10 \downarrow}$$

Across countries, we should observe a negative relationship.

Testing the ORS Effect: Cross-Country Evidence

- We compare the estimated ToT persistence, ρ_i , with the trade-balance impact response across countries.
- Under ORS, the fitted relationship should be **negative**.

Additional ORS Tests



Result

We find **no clear negative relationship** between ToT persistence and the trade-balance impact response.

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Terms of Trade as Drivers of Business Cycles

A Broader Question

How important are Terms-of-Trade shocks as **drivers of business cycles**?

- We extend the previous empirical analysis from **2 to 5 variables**.

$$\begin{aligned}x_t^1 &= \rho x_{t-1}^1 + u_t^1, \\x_t^2 &= \alpha_0 x_t^1 + \alpha_1 x_{t-1}^1 + \rho_2 x_{t-1}^2 + u_t^2,\end{aligned}$$

where

$$x_t^1 = \widehat{tot}_t, \quad x_t^2 = \begin{bmatrix} \widehat{tb}_t \\ \widehat{y}_t \\ \widehat{c}_t \\ \widehat{i}_t \end{bmatrix}.$$

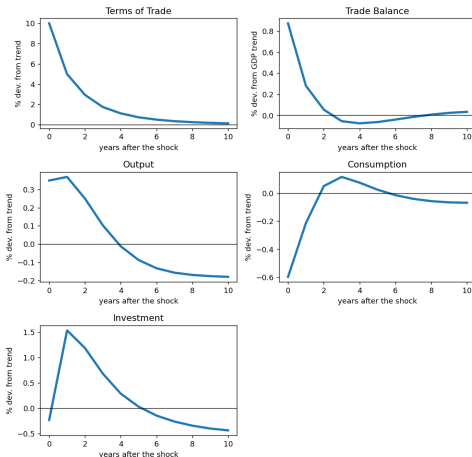
- As before, variables are detrended using a **quadratic trend**.
- ToT shocks are identified as the only structural innovation affecting ToT contemporaneously.

Macroeconomic Responses to a ToT Shock

Following a **10% improvement in the ToT**:

- **Trade balance improves**
Higher purchasing power initially raises external saving.
- **Output initially expands**
Higher real income stimulates economic activity.
- **Consumption falls on impact**
Part of the initial income windfall is saved; consumption then recovers.
- **Investment initially falls, then expands**
The response is delayed, peaking one year after the shock.

IRF a +10% no tot — quadratic



A positive ToT shock generates an initial **income and purchasing-power gain**, which propagates through saving, consumption, output, and investment.

How Important Are ToT Shocks for Business Cycles?

- Impulse responses show **how** the economy reacts to ToT shocks. To assess their quantitative importance, we compute the fraction of each variable's variance attributable to ToT shocks:

$$\text{Variance Share}_j = \frac{\text{Var}(x_{j,t} \mid \varepsilon_t^{\text{ToT}})}{\text{Var}(x_{j,t})} \times 100.$$

	ToT	Trade Balance	Output	Consumption	Investment
Cross-Country Mean	100.0	17.6	11.6	13.5	12.5
Cross-Country Median	100.0	12.2	9.8	10.6	10.2
Median Absolute Deviation	0.0	11.4	8.8	9.1	8.3
Using Mean of h_x and Π	100.0	7.0	2.9	1.5	2.2
Panel Estimation	100.0	4.0	1.3	1.2	0.5

- Values may differ slightly from those reported in the book because we use the **non-rounded estimates** in our calculations.
- Using the cross-country median, ToT shocks explain about **10–12%** of fluctuations in the main macroeconomic aggregates.

Main Result

Terms-of-Trade shocks are a **relevant but limited** source of business-cycle fluctuations in emerging and poor economies.

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The MX Model: Why Do We Need It?

- So far, the empirical SVAR has shown how the economy **responds to Terms-of-Trade shocks**.
- But the previous one-good framework treats the ToT shock essentially as a productivity shock.
- This implies an unrealistic economy in which:
 - ▶ virtually all domestic production is exportable;
 - ▶ virtually all domestic absorption is imported.
- In the data, however, countries both **produce and consume** importable and exportable goods.

Key Innovation of the MX Model

The economy explicitly contains two sectors: **Importables** (m) and **Exportables** (x).

Empirical SVAR $\implies$ Observed responses

MX Model $\implies$ Structural mechanism

The MX Model: Households

Households choose consumption, sector-specific labor, investment, capital accumulation, and external debt:

$$\max E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, h_t^m, h_t^x)$$

subject to

$$\begin{aligned} c_t + i_t^m + i_t^x + \Phi_m(k_{t+1}^m - k_t^m) + \Phi_x(k_{t+1}^x - k_t^x) + d_t \\ = \frac{d_{t+1}}{1+r_t} + w_t^m h_t^m + w_t^x h_t^x + u_t^m k_t^m + u_t^x k_t^x. \end{aligned}$$

Capital evolves separately in each sector:

$$k_{t+1}^j = (1 - \delta)k_t^j + i_t^j, \quad j \in \{m, x\}.$$

- Labor and capital are **sector-specific**.
- Adjustment costs prevent instantaneous reallocation across sectors.
- Households can borrow or lend internationally.

The MX Model: Production Structure

Sectoral production

$$y_t^m = F^m(k_t^m, h_t^m), \quad y_t^x = F^x(k_t^x, h_t^x).$$

Firms maximize profits:

$$p_t^j F^j(k_t^j, h_t^j) - w_t^j h_t^j - u_t^j k_t^j,$$

implying

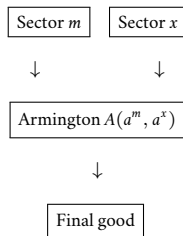
$$p_t^j F_1^j = u_t^j, \quad p_t^j F_2^j = w_t^j.$$

Final-good production

$$A(a_t^m, a_t^x)$$

combines importable and exportable goods using an **Armington aggregator**.

$$A_1(a_t^m, a_t^x) = p_t^m, \quad A_2(a_t^m, a_t^x) = p_t^x.$$



Market Clearing and the Terms-of-Trade Shock

Domestic final-good market clearing:

$$c_t + i_t^m + i_t^x + \Phi_m(\Delta k_t^m) + \Phi_x(\Delta k_t^x) = A(a_t^m, a_t^x).$$

Imports and exports are

$$m_t = p_t^m (a_t^m - y_t^m), \quad x_t = p_t^x (y_t^x - a_t^x).$$

External debt evolves according to

$$\frac{d_{t+1}}{1 + r_t} = d_t + m_t - x_t.$$

Terms of Trade

$$tot_t = \frac{p_t^x}{p_t^m}, \quad \widehat{tot}_t = \rho \widehat{tot}_{t-1} + \pi \varepsilon_t^{tot}.$$

- As in the SVAR, the economy is small and takes the ToT as **exogenous**.
- Importantly, the same estimated ρ and π from the empirical exercise discipline the model.

From the Model to the Data

- With multiple goods, model variables cannot be compared directly with the empirical aggregates used in the SVAR.
- The WDI measures real GDP using a **Paasche implicit price deflator**.
- Therefore, the model constructs theoretical counterparts consistent with the empirical measures.

Nominal GDP is

$$P_t^m y_t^m + P_t^x y_t^x.$$

Using the Paasche deflator, model-consistent real GDP becomes

$$GDP_t^{real} = p^m y_t^m + p^x y_t^x$$

where steady-state relative prices are used as base-period prices.

Why This Matters

The MX model and the empirical SVAR must measure **the same economic objects** before their predictions can be compared.

Functional Forms and Economic Frictions

Preferences: GHH

$$U(c, h^m, h^x) = \frac{[c - G(h^m, h^x)]^{1-\sigma} - 1}{1-\sigma}.$$

Sectoral labor

$$G(h^m, h^x) = \frac{(h^m)^{\omega_m}}{\omega_m} + \frac{(h^x)^{\omega_x}}{\omega_x}.$$

Production: Cobb–Douglas

$$F^j(k^j, h^j) = A^j (k^j)^{\alpha_j} (h^j)^{1-\alpha_j}.$$

Capital adjustment costs

$$\Phi_j(z) = \frac{\phi_j}{2} z^2.$$

Debt-elastic premium

$$p(d) = \bar{p} + \psi(e^{d-\bar{d}} - 1).$$

Armington aggregator

$$A(a^m, a^x) = [\chi(a^m)^{1-1/\mu} + (1-\chi)(a^x)^{1-1/\mu}]^{\frac{1}{1-1/\mu}}.$$

- ϕ_m, ϕ_x : slow capital reallocation.
- ω_m, ω_x : imperfect labor mobility.
- μ : substitution between importables and exportables.
- ψ : stabilizes external debt.

Parameterization of the MX Model

The model combines **calibration** and **country-level estimation**.

❶ Parameters inherited from standard SOE models

$$\sigma, \delta, r^*, \bar{p}, \alpha_m, \alpha_x, \omega_m, \omega_x.$$

❷ Parameters disciplined by sectoral trade data

$$\mu, \chi, \bar{d}, A_x.$$

They match average trade-balance, export, and exportable-output shares.

❸ Country-specific parameters

$$\rho, \pi, \phi_m, \phi_x, \psi.$$

Here, ρ and π come directly from the **estimated ToT process**.

	ρ	π	ϕ_m	ϕ_x	ψ
Median	0.53	0.09	1.82	1.56	0.18

Connection to the Empirical Exercise

The estimated ToT process from the first activity **feeds directly into the structural MX model**.

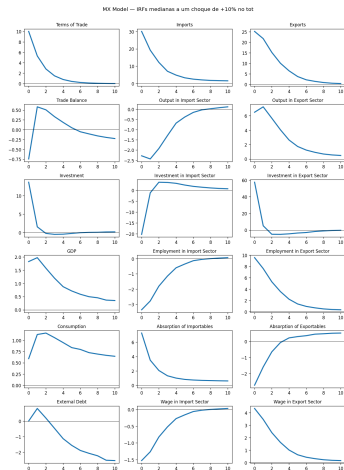
What Does the MX Model Predict?

Following a positive ToT shock:

- production shifts from **importables** to **exportables**;
- imports rise as domestic demand for importables increases;
- exports rise as exportable production expands;
- GDP and consumption increase;
- the trade balance **deteriorates on impact**.

Key Tension

The model fails to reproduce the empirical **HLM effect**: in the SVAR, the trade balance improves on impact.



MX model, cross-country median IRFs to a +10% ToT shock (replication).

Variance Explained by ToT Shocks: MX vs. SVAR

Cross-country median of the fraction of each variable's variance explained by terms-of-trade shocks (Table 7.6):

Variance share (%)	MX Model	Empirical SVAR
Terms of Trade	100.0	100.0
Trade Balance	26.9	12.2
Output	18.0	9.8
Consumption	24.1	10.6
Investment	20.3	10.2

Main Result

The MX model attributes roughly **twice as much** business-cycle variation to ToT shocks as the empirical SVAR.

Country-Level Variance Shares (Table 7.7)

Fraction of each variable's variance explained by ToT shocks (%), country by country: MX model vs. empirical SVAR. Ratios can exceed 100 when the model predicts more ToT-driven variance than the data display.

Country	Trade Bal.		Output		Consump.		Invest.	
	MX	SVAR	MX	SVAR	MX	SVAR	MX	SVAR
Argentina	4.1	25.3	5.0	30.5	5.4	17.0	5.0	25.9
Bolivia	12.5	2.5	17.9	3.5	59.6	10.6	45.1	7.6
Botswana	9.7	27.7	16.8	48.6	6.0	36.8	11.1	39.0
Brazil	60.4	43.4	15.2	10.9	14.4	3.7	19.6	16.2
El Salvador	20.0	10.7	19.9	10.3	22.3	10.2	23.6	10.7
Guatemala	39.1	5.4	18.1	2.5	12.1	3.4	30.4	4.8
Uruguay	4.1	17.3	8.3	34.3	6.2	32.9	3.8	14.4
Venezuela	26.9	41.9	15.1	23.8	9.5	33.6	8.3	11.0
Zimbabwe	3.2	7.0	4.7	10.3	7.2	33.8	2.2	16.9
Median (all 51)	26.9	12.2	18.0	9.8	24.1	10.6	20.3	10.2

Reading the Table

Considerable heterogeneity across countries; the median (Table 7.6) is the robust summary. Full 51-country table in `table_7_7_bycountry.csv`.

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Motivation: Does the Detrending Method Matter?

- To implement this extension, we had to return to the **raw data** and reconstruct the empirical analysis from the beginning.
- This made the replication more demanding, but also allowed us to reproduce the analysis under **three alternative specifications**:

Quadratic Trend (Baseline)

HP Filter

Hamilton Filter

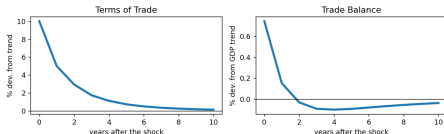
What do we want to know?

- 1 Does the relationship between **ToT shocks and the trade balance** change across detrending methods?
- 2 Does the **ORS test** remain inconclusive?
- 3 Do **macroeconomic variables** respond differently to ToT shocks?
- 4 Do ToT shocks explain **more or less of business-cycle fluctuations**?

Robustness: Does Detrending Matter?

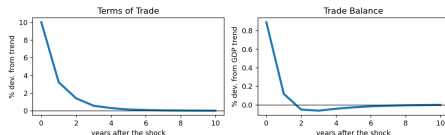
Quadratic (Baseline)

IRF a +10% no tot — quadratic



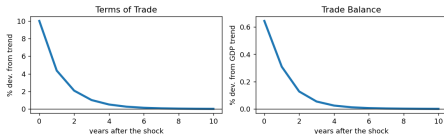
HP Filter

IRF a +10% no tot — hp



Hamilton

IRF a +10% no tot — hamilton



What Changes?

HP Filter

- Very similar to the baseline.
- Same short-lived improvement in the trade balance.

Hamilton Filter

- The response becomes more persistent.
- No short-run trade-balance reversal.
- ToT shocks are approximately **2× more volatile**.

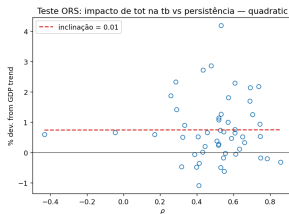
Robustness: Does the ORS Result Depend on Detrending?

ORS Prediction

More persistent ToT shocks should generate a **smaller** trade-balance response:

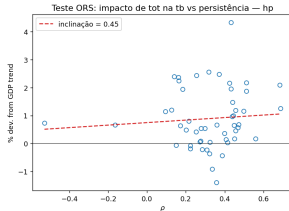
$$\rho_i \uparrow \implies IR_i^{TB}(0) \downarrow.$$

Quadratic



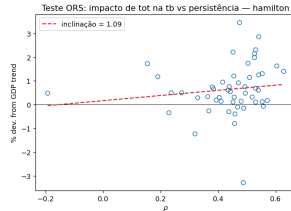
$$\hat{\beta} = 0.01$$

HP Filter



$$\hat{\beta} = 0.45$$

Hamilton



$$\hat{\beta} = 1.09$$

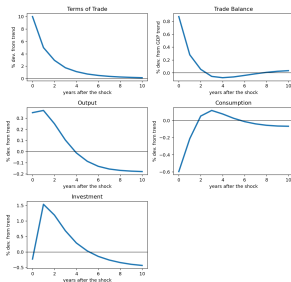
Main Result

The ORS prediction is **not recovered under any detrending method**. All estimated slopes are non-negative, and the relationship becomes more positive under HP and Hamilton detrending.

Robustness: Macroeconomic Responses Across Detrending Methods

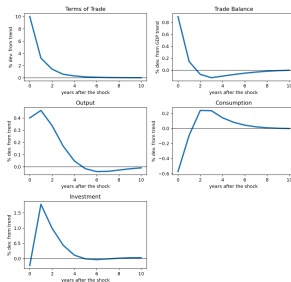
Quadratic (Baseline)

IRF a = 10% no tot — quadratic



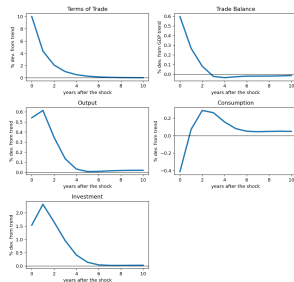
HP Filter

IRF a = 10% no tot — hp



Hamilton Filter

IRF a = 10% no tot — hamilton



Takeaway

The main qualitative results are **robust to the HP filter**. The **Hamilton filter**, however, produces noticeably stronger macroeconomic responses, consistent with the larger volatility of the identified ToT shocks.

Robustness: How Much Do ToT Shocks Explain?

- We recompute the variance contribution of ToT shocks under each detrending method.
- We focus on the **cross-country median** to compare the quantitative importance of ToT shocks.

	Trade Balance	Output	Consumption	Investment
Quadratic (Baseline)	12.2	9.8	10.6	10.2
HP Filter	13.2	12.9	14.7	14.1
Hamilton Filter	13.6	11.3	11.3	10.5

- **Trade Balance:** remarkably stable across methods, at around 12–14%.
- **HP Filter:** increases the contribution of ToT shocks, especially for consumption and investment.
- **Hamilton Filter:** remains relatively close to the quadratic baseline in terms of variance explained.

Takeaway

The conclusion is broadly robust: ToT shocks explain a **meaningful but limited share** of business-cycle fluctuations, although their quantitative importance is somewhat larger under the HP filter.

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Thank You!

Questions? Comments? Suggestions?

Appendix: Harberger–Laursen–Metzler Effect

Environment

A static small open economy with Keynesian consumption and import demand:

$$c_t = \bar{c} + \alpha y_t, \quad m_t = \mu y_t, \quad x_t = q \text{ tot}_t,$$

where $0 < \alpha < 1$, $\mu > 0$, and q is the volume of exports.

Goods-market equilibrium

$$y_t = c_t + g + i + x_t - m_t \implies y_t = \frac{\bar{c} + g + i + q \text{ tot}_t}{1 + \mu - \alpha}.$$

Using $tb_t = x_t - m_t$, the equilibrium trade balance is

$$tb_t = \frac{1 - \alpha}{1 + \mu - \alpha} q \text{ tot}_t - \frac{\mu(\bar{c} + g + i)}{1 + \mu - \alpha}$$

and therefore

$$\frac{\partial tb_t}{\partial \text{tot}_t} = \frac{1 - \alpha}{1 + \mu - \alpha} q > 0$$

Appendix: Obstfeld–Svensson–Razin Effect

Environment

A small open endowment economy in which households smooth consumption:

$$\max_{\{c_t, d_t\}} E_0 \sum_{t=0}^{\infty} \beta^t \left[-\frac{1}{2} (c_t - \bar{c})^2 \right]$$

subject to

$$d_t = (1 + r)d_{t-1} + c_t - tot_t, \quad tot_t = \rho tot_{t-1} + \varepsilon_t^{tot}.$$

Consumption smoothing implies that the trade balance responds according to

$$tb_t = rd_{t-1} + \underbrace{\frac{1 - \rho}{1 + r - \rho}}_{\text{response to ToT}} tot_t$$

Hence,

$$\frac{\partial tb_t}{\partial tot_t} = \frac{1 - \rho}{1 + r - \rho} > 0,$$

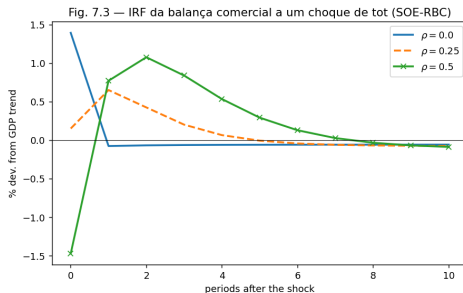
$$\frac{\partial^2 tb_t}{\partial tot_t \partial \rho} = -\frac{r}{(1 + r - \rho)^2} < 0$$

Appendix: ORS – Persistence and Trade-Balance Response

ORS Prediction

More persistent ToT shocks should generate a **smaller trade-balance response**:

$$\rho \uparrow \implies IR^{TB}(0) \downarrow$$



Result

The model illustrates how the trade-balance response changes with the persistence of the ToT shock.

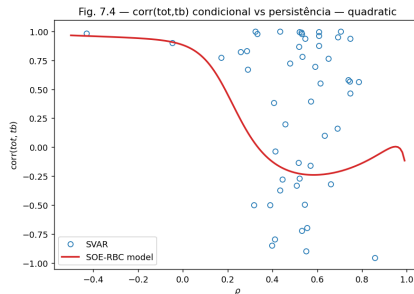
Note: Results using the HP and Hamilton filters are also available in the replication folder.

Appendix: ORS – Conditional Correlations

Alternative ORS Prediction

The SOE–RBC model links ToT persistence to the conditional correlation between ToT and the trade balance:

$$\rho \uparrow \implies \text{corr}(tot, tb \mid \varepsilon^{tot}) \downarrow$$



Result

The SVAR estimates show **no systematic relationship** matching the model prediction.

Note: Results using the HP and Hamilton filters are also available in the replication folder.